

12-Week Practice Plan — Class 5 Mathematics Olympiad

Built around the SOF IMO pattern (50 Q / 60 marks / 60 min — see [02-exam-pattern.md](#)). Same three-phase shape as the Class 4 plan: build core skills topic by topic, add pure logical reasoning, then stress-test with full timed papers — with one extra core week, since Class 5 adds two genuinely new topic areas over Class 4.

Suggested pace: **3 short sessions a week, ~30–40 minutes each.**

Phase 1 — Core Skills (Weeks 1–8)

Week	Topic	Folder
1	Number Sense & Numerals (7–8 digit numbers, lakhs/crores, Roman numerals, rounding)	assignments/01-number-sense/
2	Four Operations (large numbers, multi-digit, word problems)	assignments/02-four-operations/
3	Factors, Multiples, HCF/LCM & Number Patterns	assignments/03-factors-multiples-patterns/
4	Fractions & Decimals	assignments/04-fractions-decimals/
5	Percentage, Ratio & Proportion, Profit/Loss & Simple Interest (new vs. Class 4)	assignments/05-percentage-ratio-profit-loss/
6	Measurement (Length, Weight, Capacity, Time, Temperature) & Money	assignments/06-measurement-money/
7	Geometry, Symmetry, Perimeter & Area	assignments/07-geometry-perimeter-area/
8	Data Handling	assignments/08-data-handling/

Weekly routine: attempt the assignment untimed first; mark it against the answer key; redo only the missed questions 2 days later.

Phase 2 — Logical Reasoning (Weeks 9–10)

Week	Topic	Folder
9	Patterns, Analogy & Classification, Alphabet Test, Coding–Decoding, Logical Sequence of Words, Puzzle Test	assignments/09-reasoning-patterns-coding/ ✓
10	Ranking Test, Direction Sense, Mirror & Water Images, Embedded Figures, Clock & Calendar	assignments/10-reasoning-direction-figures/ ✓ (18 Q — Part A text-based, Part B diagram-based with SVG figures)

Phase 3 — Achievers / HOTS + Full Paper (Weeks 11–12)

Week	Activity
11	assignments/11-achievers-hots/ ✓ — 2-and-3-step mixed problems pulling from every Phase-1 topic. Then attempt Sample Paper 1 (untimed, open-book) to find the gaps — see sample-papers/README.md .
12	Attempt Sample Papers 2–5 (Level-1), each strictly timed to 60 minutes , one every 1–2 days. After each paper, log every wrong answer in an error log (topic + why it was wrong) before moving to the next paper.
13 (if qualified)	Work through the 4 Level-2 topic assignments (assignments/12-achievers-level2/ , 13-reasoning-level2/ , 14-percentage-ratio-level2/ , 15-geometry-level2/) untimed first, to rebuild each skill at the harder difficulty. Then attempt Sample Papers 6–9 (Level-2 track), timed, once Level 1 results come in and you've qualified for Level 2 — see below.

Sample paper schedule (Phase 3)

Paper	When	Mode
Sample Paper 1	Week 11, end	Untimed, review-focused
Sample Paper 2	Week 12, Day 1	Timed (60 min)
Sample Paper 3	Week 12, Day 3	Timed (60 min)
Sample Paper 4	Week 12, Day 5	Timed (60 min)
Sample Paper 5	Week 12, Day 7	Timed (60 min) — treat as Level-1 dress rehearsal
Sample Paper 6	After qualifying Level 1	Timed (60 min), Level-2 difficulty
Sample Paper 7	1–2 days after Paper 6	Timed (60 min), Level-2 difficulty
Sample Paper 8	1–2 days after Paper 7	Timed (60 min), Level-2 difficulty
Sample Paper 9	1–2 days after Paper 8	Timed (60 min), Level-2 difficulty

Level-2 topic assignments (new)

Before the Level-2 sample papers, four dedicated harder topic worksheets are available, for the areas where Level 2 gets conceptually harder (not just bigger numbers):

Folder	Topic
assignments/12-achievers-level2/ 	Multi-step HOTS (chained percentage changes, fraction-of-remainder, age/rate problems)
assignments/13-reasoning-level2/ 	Compound logical reasoning (2+ stacked clues per question)
assignments/14-percentage-ratio-level2/ 	Multi-step percentage/ratio/profit-loss/SI — the topic newest to Class 5
assignments/15-geometry-level2/ 	Compound-shape perimeter & area (paths, frames, cut corners)

All 9 sample papers are built out (full 50 questions, 60 marks, answer key each) — see [sample-papers/README.md](#). Papers 1–5 are Level-1 difficulty; Papers 6–9 are a dedicated **Level-2 track** for students who've qualified Level 1 and are prepping for the actual, harder Level-2 exam.

Progress tracker

- ☐ Week 1 — Number Sense (*assignment ready*)
- ☐ Week 2 — Four Operations (*assignment ready*)
- ☐ Week 3 — Factors, Multiples, HCF/LCM & Patterns (*assignment ready*)
- ☐ Week 4 — Fractions & Decimals (*assignment ready*)
- ☐ Week 5 — Percentage, Ratio & Proportion, Profit/Loss & Simple Interest (*assignment ready*)
- ☐ Week 6 — Measurement & Money (*assignment ready*)
- ☐ Week 7 — Geometry, Perimeter & Area (*assignment ready*)
- ☐ Week 8 — Data Handling (*assignment ready*)
- ☐ Week 9 — Reasoning: Patterns/Analogy/Coding/Word Sequence/Puzzles (*assignment ready*)
- ☐ Week 10 — Reasoning: Ranking/Direction/Mirror & Water Images (*assignment ready*)
- ☐ Week 11 — Achievers/HOTS + Sample Paper 1 (*assignment + paper ready*)
- ☐ Week 12 — Sample Papers 2–5, timed (*Level-1, all ready*)
- ☐ Week 13 — Level-2 topic assignments (Achievers, Reasoning, Percentage/Ratio, Geometry), if qualified (*all ready*)
- ☐ Week 13 — Sample Papers 6–9, timed, if qualified for Level 2 (*Level-2 track, all ready*)

(The checkboxes above track *your* progress working through the plan, not whether the content exists — every assignment and sample paper listed is already built.)

Optional extension

Once the core plan is comfortably done, the Class 4 kit's [Math Kangaroo-style stretch set](#) is equally suitable here — Math Kangaroo's grade bands overlap (the "Ecolier" level spans grades 3–4, "Benjamin" spans grades 5–6), so a Class 5 student is better matched to the *next* band up in puzzle difficulty rather than that exact set. Worth building a Class-5-pitched version of it as a future addition.